

TYPICAL SMOKE DETECTOR OBSCURATION RATINGS

Type	Rating
Photoelectric	6.5% obs/m to 13.0% obs/m (2% obs/ft to 4% obs/ft)
Ionisation	2.6% obs/m to 5.0% obs/m (0.8% obs/ft to 1.5% obs/ft)
Aspiration	0.005% obs/m to 20.5% obs/m (0.0015% obs/ft to 6.25% obs/ft)
Laser	0.06% obs/m to 6.41% obs/m (0.02% obs/ft to 2.0% obs/ft)

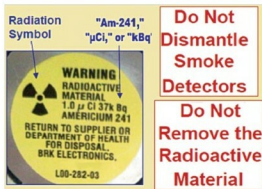


Fig. 6: Warning labels on an ionisation smoke detector

the immediate vicinity rises higher than the pre-set rate-per-time factor.

What to look for

Photoelectric (optical) smoke detectors are quicker in detecting smoke generated by smoldering (smoke) fires. Ionisation smoke detectors are quicker in detecting smoke generated by flaming (hot) fires.

Obscuration is a unit of measurement that has become the standard definition of smoke detector sensitivity. It is the effect

that smoke has on reducing visibility (higher concentrations of smoke result in higher obscuration levels, lowering visibility). Typical smoke detector obscuration rating is shown in the table.

For each type of smoke alarm, the advantage it provides may be critical to life safety in some fire situations. Fatal fires include a large number of smoldering fires and a large number of flaming fires. As it is very difficult to predict the type of fire disaster, or when it will occur, in addition to individual photoelectric and ionisation smoke detectors, a combination of detectors that include multiple technologies in a single unit are recommended for reliable protection. Any smoke detector technology, to be acceptable, must perform tolerably for all types of fires in order to provide early warning of fire at all times of the day or night. **EFY**

air temperature exceeds the pre-set temperature. Rate-of-rise detectors raise an alert when temperature in

Simulating Environmental Conditions for Electronics Testing



Climatic Chamber

- -70°C to 180°C, 10%RH to 98%RH
- Perform Cyclic Low, High Temperature and Humidity Test.
- Meets test standards such as ASTM, IEC, MIL, DIN, ISO, JIS etc.



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